Solutions Tutorial 1

Exercise 1

(a) UMP:

$$\max_{x \geq 0} (x_1 - b_1)^{\alpha}(x_2 - b_2)^{1-\alpha} \qquad \text{s.t.} \qquad p \cdot x \leq w.$$

(b) Let us maximise the Lagrangian while assuming an interior solution:

$$\max_{x \geq 0,\, \lambda} \mathcal{L} = (x_1 - b_1)^{\alpha}(x_2 - b_2)^{1-\alpha} + \lambda\,(w - p_1 x_1 - p_2 x_2).$$

First-order conditions:

$$\frac{\partial \mathcal{L}}{\partial x_1} = \alpha\,(x_1 - b_1)^{\alpha-1}(x_2 - b_2)^{1-\alpha} - \lambda p_1 = 0 \tag{1}$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = (1-\alpha)\,(x_1 - b_1)^{\alpha}(x_2 - b_2)^{-\alpha} - \lambda p_2 = 0 \tag{2}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = w - p_1 x_1 - p_2 x_2 = 0 \tag{3}$$

Dividing (1) by (2):

$$\frac{\alpha\,(x_2 - b_2)}{(1-\alpha)(x_1 - b_1)} = \frac{p_1}{p_2} \quad \Longrightarrow \quad x_2 = \frac{1-\alpha}{\alpha} \cdot \frac{p_1}{p_2}\,(x_1 - b_1) + b_2. \tag{a}$$

From (3):

$$x_1 = \frac{w - p_2 x_2}{p_1}. \tag{b}$$

Substitute (b) in (a):

$$x_2 = \frac{1-\alpha}{\alpha\, p_2}\left(w - p_2 x_2 - p_1 b_1\right) + b_2 \quad \Longleftrightarrow \quad x_2\Big(1 + \tfrac{1-\alpha}{\alpha}\Big) = \frac{x_2}{\alpha} = \frac{1-\alpha}{\alpha\, p_2}\,(w - p_1 b_1) + b_2,$$

so that

$$x_2 = \frac{1-\alpha}{p_2}\left[w - p_1 b_1\right] + \alpha\, b_2.$$

Writing $\alpha b_2 = b_2 - (1-\alpha)b_2 = b_2 - \frac{1-\alpha}{p_2}\, p_2 b_2$, this becomes

$$\boxed{x_2(p, w) = \frac{1-\alpha}{p_2}\left[w - p_1 b_1 - p_2 b_2\right] + b_2} \tag{c}$$

Substitute (c) in (b) and rewrite:

$$\boxed{x_1(p, w) = \frac{\alpha}{p_1}\left[w - p_1 b_1 - p_2 b_2\right] + b_1}$$

One usually rewrites demand as follows:

$$p_1 x_1(p, w) = \alpha\left[w - p_1 b_1 - p_2 b_2\right] + p_1 b_1$$
$$p_2 x_2(p, w) = (1-\alpha)\left[w - p_1 b_1 - p_2 b_2\right] + p_2 b_2$$

This is known as the Linear Expenditure System (LES): the consumer first buys the subsistence bundle (b_1, b_2) and then spends the fractions α and $1-\alpha$ of the remaining ("supernumerary") income $w - p_1 b_1 - p_2 b_2$ on goods 1 and 2.

(c) Substitute $x_1(p,w)$ and $x_2(p,w)$ in $u(x)$ to obtain the indirect utility function:

$$v(p,w) = \left[\frac{\alpha}{p_1}\left(w - p_1 b_1 - p_2 b_2\right)\right]^{\alpha} \left[\frac{1-\alpha}{p_2}\left(w - p_1 b_1 - p_2 b_2\right)\right]^{1-\alpha}$$

$$v(p,w) = \left(\frac{\alpha}{p_1}\right)^{\alpha} \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} \left[w - p_1 b_1 - p_2 b_2\right]$$

(d) (i) Homogeneity of degree zero: we need $x(\gamma p, \gamma w) = x(p,w)$ for all $\gamma > 0$.

$$x_1(\gamma p, \gamma w) = \frac{\alpha}{\gamma p_1}\left[\gamma w - \gamma p_1 b_1 - \gamma p_2 b_2\right] + b_1 = \frac{\alpha}{p_1}\left[w - p_1 b_1 - p_2 b_2\right] + b_1 = x_1(p,w).$$

Similarly for the second commodity.

(ii) Adding-up (Walras' law): we need $p_1 x_1 + p_2 x_2 = w$.

$$\begin{aligned} p_1 x_1 + p_2 x_2 &= \alpha\left(w - p_1 b_1 - p_2 b_2\right) + p_1 b_1 + (1-\alpha)\left(w - p_1 b_1 - p_2 b_2\right) + p_2 b_2 \\ &= w - p_1 b_1 - p_2 b_2 + p_1 b_1 + p_2 b_2 = w. \end{aligned}$$

(e) (i) Homogeneity of degree zero of the indirect utility function: we need $v(\gamma p, \gamma w) = v(p,w)$ for any $\gamma > 0$.

$$\begin{aligned} v(\gamma p, \gamma w) &= \left(\frac{\alpha}{\gamma p_1}\right)^{\alpha} \left(\frac{1-\alpha}{\gamma p_2}\right)^{1-\alpha} \left[\gamma w - \gamma p_1 b_1 - \gamma p_2 b_2\right] \\ &= \left(\frac{\alpha}{p_1}\right)^{\alpha} \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} \underbrace{\left(\frac{1}{\gamma}\right)^{\alpha} \left(\frac{1}{\gamma}\right)^{1-\alpha} \gamma}_{=1} \left[w - p_1 b_1 - p_2 b_2\right] = v(p,w). \end{aligned}$$

(ii)

$$\frac{\partial v(p,w)}{\partial w} = \left(\frac{\alpha}{p_1}\right)^{\alpha} \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} > 0.$$

Since $\alpha \in (0,1)$ and prices are strictly positive, the indirect utility function is strictly increasing in w.

(iii)

$$\frac{\partial v(p,w)}{\partial p_1} = \underbrace{\alpha\left(\frac{\alpha}{p_1}\right)^{\alpha-1}}_{>0} \underbrace{\left(-\frac{\alpha}{p_1^2}\right)}_{<0} \underbrace{\left(\frac{1-\alpha}{p_2}\right)^{1-\alpha}}_{>0} \underbrace{\left[w - p_1 b_1 - p_2 b_2\right]}_{>0} + \underbrace{\left(\frac{\alpha}{p_1}\right)^{\alpha} \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha}}_{>0} \underbrace{(-b_1)}_{<0}$$

Both terms are negative (note that $w - p_1 b_1 - p_2 b_2 > 0$ follows from $x_1 > b_1$ and $x_2 > b_2$), so

$$\frac{\partial v(p,w)}{\partial p_1} < 0.$$

Remark: collecting terms,

$$\frac{\partial v(p,w)}{\partial p_1} = -\left(\frac{\alpha}{p_1}\right)^{\alpha} \left(\frac{1-\alpha}{p_2}\right)^{1-\alpha} \left[\frac{\alpha}{p_1}\left(w - p_1 b_1 - p_2 b_2\right) + b_1\right] = -\frac{\partial v(p,w)}{\partial w} x_1(p,w),$$

which is Roy's identity (see Tutorial 2).

(f) Solve the new UMP where $u(x)$ is replaced by $\tilde{u}(x)$:

$$\max_{x_1, x_2, \lambda} \tilde{\mathcal{L}} = \alpha \ln(x_1 - b_1) + (1-\alpha)\ln(x_2 - b_2) + \lambda\left(w - p_1 x_1 - p_2 x_2\right).$$

The first-order conditions are

$$\frac{\alpha}{x_1 - b_1} = \lambda p_1, \qquad \frac{1-\alpha}{x_2 - b_2} = \lambda p_2, \qquad w - p_1 x_1 - p_2 x_2 = 0.$$

Dividing the first two gives $\dfrac{\alpha\,(x_2 - b_2)}{(1-\alpha)(x_1 - b_1)} = \dfrac{p_1}{p_2}$, i.e. exactly condition (a) above. This results in exactly the same demand equations as before.

Why? $\tilde{u}(x) = \ln u(x)$ is a positive monotone transformation of $u(x)$, so it represents the same preferences and therefore generates the same demand.

Exercise 2

(a) The budget share of good ℓ is

$$s_\ell(p, w) = \frac{p_\ell\, x_\ell(p, w)}{w}.$$

Let us take the first derivative of this budget share with respect to w:

$$\begin{aligned}
\frac{\partial s_\ell(p,w)}{\partial w} &= \frac{p_\ell \dfrac{\partial x_\ell(p,w)}{\partial w}\, w - p_\ell\, x_\ell(p,w)}{w^2} \\
&= \frac{p_\ell\, x_\ell(p,w)}{w^2} \cdot \frac{\partial x_\ell(p,w)}{\partial w} \cdot \frac{w}{x_\ell(p,w)} - \frac{p_\ell\, x_\ell(p,w)}{w^2} \\
&= \frac{s_\ell(p,w)}{w}\, \varepsilon_{\ell w}(p,w) - \frac{s_\ell(p,w)}{w} = \big[\varepsilon_{\ell w}(p,w) - 1\big]\, \frac{s_\ell(p,w)}{w},
\end{aligned}$$

where $\varepsilon_{\ell w}(p,w) = \dfrac{\partial x_\ell(p,w)}{\partial w} \cdot \dfrac{w}{x_\ell(p,w)}$ is the income elasticity of good ℓ. If $\varepsilon_{\ell w}(p,w) > 1$ (good ℓ is a luxury good), then $\dfrac{\partial s_\ell(p,w)}{\partial w} > 0$: the budget share is increasing in w.

(b) Recall

$$x_1(p,w) = \frac{\alpha}{p_1}\big[w - p_1 b_1 - p_2 b_2\big] + b_1, \qquad x_2(p,w) = \frac{1-\alpha}{p_2}\big[w - p_1 b_1 - p_2 b_2\big] + b_2.$$

Commodity 1: $\dfrac{\partial x_1(p,w)}{\partial w} = \dfrac{\alpha}{p_1}$, so

$$\varepsilon_{1w}(p,w) = \frac{\alpha}{p_1} \cdot \frac{w}{\dfrac{\alpha}{p_1}\big[w - p_1 b_1 - p_2 b_2\big] + b_1}.$$

Commodity 2:

$$\varepsilon_{2w}(p,w) = \frac{1-\alpha}{p_2} \cdot \frac{w}{\dfrac{1-\alpha}{p_2}\big[w - p_1 b_1 - p_2 b_2\big] + b_2}.$$

For the given values ($w = 1000$, $p_1 = 5$, $p_2 = 1$, $\alpha = 0.3$, $b_1 = 50$, $b_2 = 20$), supernumerary income is $w - p_1 b_1 - p_2 b_2 = 1000 - 250 - 20 = 730$ and the demands are

$$x_1 = \frac{0.3}{5} \cdot 730 + 50 = 93.8, \qquad x_2 = \frac{0.7}{1} \cdot 730 + 20 = 531$$

(check: $5 \times 93.8 + 1 \times 531 = 1000 = w$). Hence

$$\varepsilon_{1w}(p,w) = \frac{0.3}{5} \cdot \frac{1000}{93.8} = 0.64, \qquad \varepsilon_{2w}(p,w) = \frac{0.7}{1} \cdot \frac{1000}{531} = 1.32.$$

Good 1 is a necessity ($\varepsilon_{1w} < 1$) and good 2 is a luxury ($\varepsilon_{2w} > 1$). The budget shares are $s_1 = 469/1000 = 0.469$ and $s_2 = 0.531$; as a check, $s_1\varepsilon_{1w} + s_2\varepsilon_{2w} = 0.469 \times 0.64 + 0.531 \times 1.32 = 1$ (Engel aggregation), as it must be.

(c) The own-price elasticity is

$$\varepsilon_{\ell\ell}(p,w) = \frac{\partial x_\ell(p,w)}{\partial p_\ell} \cdot \frac{p_\ell}{x_\ell(p,w)}.$$

For good 1:

$$\frac{\partial x_1(p,w)}{\partial p_1} = -\frac{\alpha}{p_1^2}\big[w - p_1 b_1 - p_2 b_2\big] + \frac{\alpha}{p_1}(-b_1) = -\frac{\alpha}{p_1^2}\big[w - p_2 b_2\big]. \qquad (*)$$

Post-multiply by $p_1/x_1(p,w)$ and fill in the given values for w, p_1, p_2 and the parameters α, b_1, b_2:

$$\varepsilon_{11}(p,w) = -\frac{\alpha}{p_1} \cdot \frac{w - p_2 b_2}{x_1(p,w)} = -\frac{0.3}{5} \cdot \frac{980}{93.8} = -0.63.$$

Similar reasoning for good 2:

$$\frac{\partial x_2(p,w)}{\partial p_2} = -\frac{1-\alpha}{p_2^2}\big[w - p_1 b_1\big], \qquad \varepsilon_{22}(p,w) = -\frac{1-\alpha}{p_2} \cdot \frac{w - p_1 b_1}{x_2(p,w)} = -\frac{0.7}{1} \cdot \frac{750}{531} = -0.99.$$

Note: $(*)$ is always negative, so no Giffen goods are possible with Stone–Geary preferences. Both own-price elasticities are smaller than one in absolute value (demand is inelastic); this is because the subsistence part b_ℓ of demand does not react to prices at all.

(d) The cross-price elasticity is

$$\varepsilon_{12}(p,w) = \frac{\partial x_1(p,w)}{\partial p_2} \cdot \frac{p_2}{x_1(p,w)}.$$

Since $\dfrac{\partial x_1(p,w)}{\partial p_2} = \dfrac{\alpha}{p_1}(-b_2)$,

$$\varepsilon_{12}(p,w) = -\frac{\alpha\, p_2 b_2}{p_1\, x_1(p,w)} = -\frac{0.3 \times 1 \times 20}{5 \times 93.8} = -0.013.$$

Similar reasoning for ε_{21}: $\dfrac{\partial x_2(p,w)}{\partial p_1} = \dfrac{1-\alpha}{p_2}(-b_1)$, so

$$\varepsilon_{21}(p,w) = -\frac{(1-\alpha)\, p_1 b_1}{p_2\, x_2(p,w)} = -\frac{0.7 \times 5 \times 50}{1 \times 531} = -0.33.$$

The cross-price elasticities are negative: with Stone–Geary preferences the goods are gross complements, because a higher price of the other good's subsistence quantity eats into supernumerary income.

Summary for the given values:

	good 1	good 2
$x_\ell(p,w)$	93.8	531
budget share s_ℓ	0.469	0.531
income elasticity $\varepsilon_{\ell w}$	0.64	1.32
own-price elasticity $\varepsilon_{\ell\ell}$	-0.63	-0.99
cross-price elasticity $\varepsilon_{\ell k}$ $(k \neq \ell)$	-0.013	-0.33

Checks: homogeneity of degree zero implies $\varepsilon_{\ell 1} + \varepsilon_{\ell 2} + \varepsilon_{\ell w} = 0$ for each ℓ (indeed $-0.63 - 0.013 + 0.64 = 0$ and $-0.33 - 0.99 + 1.32 = 0$).

Note that the elasticities could be very different for other incomes and prices, or for other preference parameters (α, b_1 and b_2)!

(e) Suppose $b_1 = b_2 = 0$ (Cobb–Douglas). Then

$$x_1(p,w) = \frac{\alpha}{p_1}\, w \implies \text{budget share } s_1 = \alpha,$$
$$x_2(p,w) = \frac{1-\alpha}{p_2}\, w \implies \text{budget share } s_2 = 1-\alpha,$$

and

$$\begin{aligned} \text{own-price elasticities} &= -1, \\ \text{cross-price elasticities} &= 0, \\ \text{income elasticities} &= 1. \end{aligned}$$

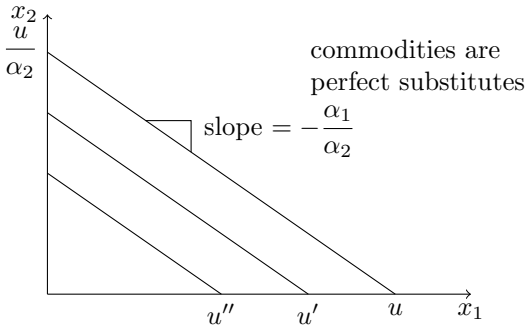
Not realistic.

Exercise 3

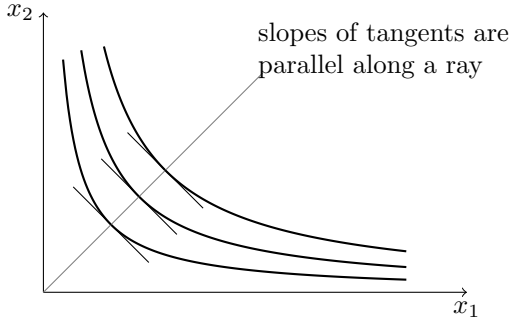
(a) $\rho = 1 \implies u(x) = \alpha_1 x_1 + \alpha_2 x_2$. Take an arbitrary utility level u (which we hold constant) and see how x_1 and x_2 are related:

$$x_2 = \frac{u}{\alpha_2} - \frac{\alpha_1}{\alpha_2}\, x_1.$$

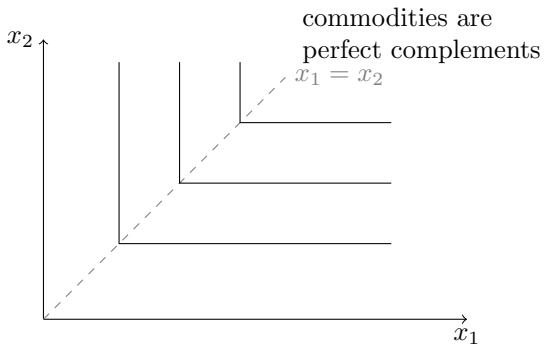
The indifference curves are straight lines with intercept u/α_2 and slope $-\alpha_1/\alpha_2$. The commodities are perfect substitutes.



(b) $\rho \to 0 \implies u(x) = x_1^{\alpha_1} x_2^{\alpha_2}$. These are Cobb–Douglas preferences. Preferences are homothetic: the slopes of the tangents to the indifference curves are parallel to each other along a given ray through the origin.



(c) $\rho \to -\infty$ (with $\alpha_1 = \alpha_2 = 1$) $\implies u(x) = \min\{x_1, x_2\}$. An indifference curve at level u consists of all bundles with $\min\{x_1, x_2\} = u$, i.e. the L-shaped set $\{x : x_1 = u,\ x_2 \geq u\} \cup \{x : x_2 = u,\ x_1 \geq u\}$, with the kink on the 45° line. This is the Leontief utility function. The commodities are perfect complements.



(d) The elasticity of substitution between goods 1 and 2 is

$$\xi_{12}(p,w) = -\frac{\partial\big[x_1(p,w)/x_2(p,w)\big]}{\partial\big[p_1/p_2\big]} \cdot \frac{p_1/p_2}{x_1(p,w)/x_2(p,w)}.$$

Assume $\alpha_1 = \alpha_2 = 1$ so that we focus on

$$u(x) = \big[x_1^\rho + x_2^\rho\big]^{1/\rho}.$$

If you set up a UMP and solve it to obtain the demand functions you get (see the Appendix at the end of this bundle):

$$x_1(p,w) = \frac{w}{p_1^\delta + p_2^\delta}\, p_1^{\delta-1}, \qquad x_2(p,w) = \frac{w}{p_1^\delta + p_2^\delta}\, p_2^{\delta-1}, \qquad \text{where } \delta = \frac{\rho}{\rho-1}.$$

Let us now calculate the elasticity of substitution. Note that

$$\frac{x_1(p,w)}{x_2(p,w)} = \left(\frac{p_1}{p_2}\right)^{\delta-1}.$$

Remark that this is a very convenient form, given that it is a function of p_1/p_2 only. Then

$$\frac{\partial\big[x_1(p,w)/x_2(p,w)\big]}{\partial\big[p_1/p_2\big]} = (\delta-1)\left(\frac{p_1}{p_2}\right)^{\delta-2},$$

so

$$\xi_{12}(p,w) = -(\delta-1)\left(\frac{p_1}{p_2}\right)^{\delta-2} \cdot \frac{(p_1/p_2)}{(p_1/p_2)^{\delta-1}} = -(\delta-1) = \frac{1}{1-\rho},$$

using $\delta - 1 = \dfrac{\rho}{\rho-1} - 1 = \dfrac{1}{\rho-1}$. We thus can derive that

$$\begin{aligned}
&\xi_{12}(p,w) = +\infty \quad \text{for the linear case } (\rho = 1),\\
&\xi_{12}(p,w) = 1 \qquad \text{for Cobb–Douglas } (\rho \to 0),\\
&\xi_{12}(p,w) = 0 \qquad \text{for Leontief } (\rho \to -\infty).
\end{aligned}$$

Exercise 4

Throughout, write

$$e_k(p,w) = p_k\, x_k(p,w)$$

for the expenditure on good k, and label the three assumptions:

(W) Walras' law: $\displaystyle\sum_{\ell=1}^{L} p_\ell\, x_\ell(p,w) = w$ for all (p,w).

(H) Homogeneity of degree zero: $x(\gamma p, \gamma w) = x(p,w)$ for all $\gamma > 0$ and all (p,w).

(N) No cross-price effects: $\dfrac{\partial x_\ell(p,w)}{\partial p_k} = 0$ for all $\ell \neq k$ and all (p,w).

Part (a) uses (W) and (N); part (b) uses (H) and (W); part (c) only collects the results.

(a) What we have to show: for each good k, the number $e_k(p,w) = p_k\, x_k(p,w)$ does not change when any price changes. It may depend on w, but not on p.

Step 1: differentiate Walras' law with respect to p_k. Assumption (W) is an identity: the left-hand side and the right-hand side are equal at *every* (p,w), not just at one point. Two functions that coincide everywhere have the same partial derivatives, so we may differentiate both sides with respect to p_k. The right-hand side is w, which does not depend on p_k, so its derivative is 0. On the left-hand side we differentiate the sum term by term:

- for a good $\ell \neq k$, the price p_ℓ does not depend on p_k, so $\dfrac{\partial\,[p_\ell\, x_\ell(p,w)]}{\partial p_k} = p_\ell\, \dfrac{\partial x_\ell(p,w)}{\partial p_k}$;
- for the good $\ell = k$ itself, p_k appears both as the price and inside $x_k(p,w)$, so by the product rule $\dfrac{\partial\,[p_k\, x_k(p,w)]}{\partial p_k} = x_k(p,w) + p_k\, \dfrac{\partial x_k(p,w)}{\partial p_k}$.

Putting the pieces together,

$$x_k(p,w) + p_k\, \frac{\partial x_k(p,w)}{\partial p_k} + \sum_{\ell \neq k} p_\ell\, \frac{\partial x_\ell(p,w)}{\partial p_k} = 0. \tag{4.1}$$

(This identity holds for *any* demand function satisfying Walras' law and is known as Cournot aggregation: after a change in p_k, the changes in expenditure on all goods must add up to zero, because total expenditure stays equal to w.)

Step 2: use the absence of cross-price effects. By (N), every derivative $\partial x_\ell/\partial p_k$ with $\ell \neq k$ in the sum of (4.1) is zero, so the whole sum vanishes and (4.1) reduces to

$$x_k(p,w) + p_k\, \frac{\partial x_k(p,w)}{\partial p_k} = 0. \tag{4.2}$$

By the product rule, the left-hand side of (4.2) is exactly $\dfrac{\partial\,[p_k\, x_k(p,w)]}{\partial p_k} = \dfrac{\partial e_k(p,w)}{\partial p_k}$. So we have shown

$$\frac{\partial e_k(p,w)}{\partial p_k} = 0 \qquad \text{at every } (p,w).$$

In words: when its own price changes, the amount spent on good k does not change.

Step 3: the other prices. Take any price p_j with $j \neq k$. When p_j varies, p_k is held fixed, so

$$\frac{\partial e_k(p,w)}{\partial p_j} = \frac{\partial\,[p_k\, x_k(p,w)]}{\partial p_j} = p_k\, \frac{\partial x_k(p,w)}{\partial p_j} = 0,$$

where the last equality is assumption (N) with the roles of the indices reversed (x_k does not react to p_j).

Step 4: from "all partial derivatives are zero" to "does not depend on p". Steps 2 and 3 together say that

$$\frac{\partial e_k(p,w)}{\partial p_j} = 0 \qquad \text{for every } j = 1,\dots,L \text{ and at every } (p,w).$$

So e_k is a function all of whose partial derivatives with respect to $p_1,\dots,p_L$ vanish everywhere. By the fact stated in the question, such a function does not depend on p: for a fixed w, $e_k(p,w)$ takes the same value at every price vector. Call that common value $g_k(w)$:

$$p_k\, x_k(p,w) = g_k(w) \qquad \text{for all } (p,w).$$

This is what we had to show.

Interpretation: if the amount spent on good k changed when p_k rose, the difference would have to be absorbed by some other good, whose demand would then depend on p_k; assumption (N) rules this out. Note also that (4.2) can be rearranged as $\dfrac{\partial x_k(p,w)}{\partial p_k}\cdot\dfrac{p_k}{x_k(p,w)} = -1$: every good has own-price elasticity $\varepsilon_{kk} = -1$.

(b) What we have to show: the function g_k found in (a) is of the form $g_k(w) = \bar{s}_k\, w$, with $\bar{s}_k \geq 0$ and $\sum_k \bar{s}_k = 1$.

Step 1: rewrite demand. Since $p_k > 0$, we can divide the conclusion of (a) by p_k:

$$x_k(p,w) = \frac{g_k(w)}{p_k}. \tag{4.3}$$

So far g_k is an arbitrary function of w; homogeneity is what will pin down its shape.

Step 2: apply homogeneity of degree zero. Take any $\gamma > 0$ and evaluate (4.3) at the scaled point $(\gamma p, \gamma w)$: the k-th price is now γp_k and wealth is γw, so

$$x_k(\gamma p, \gamma w) = \frac{g_k(\gamma w)}{\gamma\, p_k}.$$

By (H), the left-hand side equals $x_k(p,w)$, which by (4.3) is $g_k(w)/p_k$. Hence

$$\frac{g_k(\gamma w)}{\gamma\, p_k} = \frac{g_k(w)}{p_k}.$$

Multiplying both sides by $\gamma\, p_k > 0$ gives

$$g_k(\gamma w) = \gamma\, g_k(w) \qquad \text{for all } w > 0 \text{ and all } \gamma > 0. \tag{4.4}$$

So g_k is homogeneous of degree one: scaling its argument by γ scales its value by γ.

Step 3: a homogeneous-of-degree-one function of one variable is linear. Set $w = 1$ in (4.4):

$$g_k(\gamma) = \gamma\, g_k(1) \qquad \text{for all } \gamma > 0.$$

The symbol γ now plays the role of an arbitrary wealth level, so renaming it w,

$$g_k(w) = \bar{s}_k\, w \qquad \text{with } \bar{s}_k := g_k(1).$$

The graph of g_k is a straight line through the origin with slope $\bar{s}_k$.

Step 4: the sign of $\bar{s}_k$. By (a), $\bar{s}_k = g_k(1) = p_k\, x_k(p,1)$ for any price vector p. Prices are strictly positive and demand is non-negative, so $\bar{s}_k \geq 0$.

Step 5: the sum of the $\bar{s}_k$. Apply Walras' law (W) at an arbitrary (p,w) and substitute what we know about each term:

$$w \;=\; \sum_{k=1}^{L} p_k\, x_k(p,w) \;=\; \sum_{k=1}^{L} g_k(w) \;=\; \sum_{k=1}^{L} \bar{s}_k\, w \;=\; w \sum_{k=1}^{L} \bar{s}_k.$$

Dividing by $w > 0$ gives $\sum_{k=1}^{L} \bar{s}_k = 1$. This completes (b).

(c) Putting (a) and (b) together: $p_k\, x_k(p,w) = g_k(w) = \bar{s}_k\, w$, so

$$x_k(p,w) = \frac{\bar{s}_k\, w}{p_k}, \qquad k = 1, \ldots, L.$$

The budget share of good k is $s_k(p,w) = p_k\, x_k(p,w)/w = \bar{s}_k$, a constant that depends neither on prices nor on wealth. This is exactly the demand system found in Exercise 2(e) (with $L = 2$, $\bar{s}_1 = \alpha$ and $\bar{s}_2 = 1 - \alpha$), i.e. the demand generated by the Cobb–Douglas utility function $u(x) = \prod_k x_k^{\bar{s}_k}$.

What has been added: Exercise 2(e) established one direction, that Cobb–Douglas preferences *imply* zero cross-price elasticities. Here we have shown the converse: a demand function without cross-price effects (that satisfies Walras' law and is homogeneous of degree zero, as any Marshallian demand function generated by locally nonsatiated preferences does) *must* be of the Cobb–Douglas form. Taken together, the absence of cross-price effects *characterises* Cobb–Douglas demand. Two remarks. (i) The argument only used (W) and (H), never that the demand function comes from a utility maximisation problem. (ii) Along the way we found that every good has $\varepsilon_{kk} = -1$ (Step 2 of (a)) and, from $x_k = \bar{s}_k\, w/p_k$, $\varepsilon_{kw} = 1$; these are precisely the values found in Exercise 2(e).

Appendix: derivation of the demand functions associated with the CES utility function

UMP:

$$\max_{x\geq 0} \left[x_1^\rho + x_2^\rho\right]^{1/\rho} \qquad \text{s.t.} \qquad p\cdot x \leq w.$$

Let us assume that $\rho \in (-\infty, 1)\setminus\{0\}$. This will allow us to avoid the special cases with corner solutions, etc. If there is no corner solution, then we have two equations to solve for the optimal demand of x_1 and x_2:

$$\frac{\partial u(x)/\partial x_1}{\partial u(x)/\partial x_2} = \frac{p_1}{p_2} \qquad \text{and} \qquad p_1x_1 + p_2x_2 = w.$$

(Alternatively you could work with a Lagrangian.)

The tangency condition gives

$$\frac{\frac{1}{\rho}\left[x_1^\rho + x_2^\rho\right]^{\frac{1}{\rho}-1}\cdot \rho\, x_1^{\rho-1}}{\frac{1}{\rho}\left[x_1^\rho + x_2^\rho\right]^{\frac{1}{\rho}-1}\cdot \rho\, x_2^{\rho-1}} = \frac{p_1}{p_2} \quad\Longleftrightarrow\quad \frac{x_1^{\rho-1}}{x_2^{\rho-1}} = \frac{p_1}{p_2} \quad\Longleftrightarrow\quad \frac{x_1}{x_2} = \left(\frac{p_1}{p_2}\right)^{\frac{1}{\rho-1}}. \qquad (**)$$

From $(**)$ we get $x_1 = \left(\frac{p_1}{p_2}\right)^{\frac{1}{\rho-1}} x_2$. We substitute this in the budget constraint to obtain

$$p_1\left(\frac{p_1}{p_2}\right)^{\frac{1}{\rho-1}} x_2 + p_2x_2 = w \quad\Longleftrightarrow\quad \underbrace{\left[p_1\left(\frac{p_1}{p_2}\right)^{\frac{1}{\rho-1}} + p_2\right]}_{(\dagger)} x_2 = w.$$

The term in brackets can be rewritten as

$$(\dagger) = \frac{p_1^{\frac{1}{\rho-1}+1}}{p_2^{\frac{1}{\rho-1}}} + p_2 = \frac{p_1^{\frac{\rho}{\rho-1}} + p_2^{\frac{1}{\rho-1}+1}}{p_2^{\frac{1}{\rho-1}}} = \frac{p_1^{\frac{\rho}{\rho-1}} + p_2^{\frac{\rho}{\rho-1}}}{p_2^{\frac{1}{\rho-1}}} = \frac{p_1^\delta + p_2^\delta}{p_2^{\delta-1}},$$

where $\delta = \dfrac{\rho}{\rho-1}$ and $\delta - 1 = \dfrac{1}{\rho-1}$.

So we have

$$\frac{p_1^\delta + p_2^\delta}{p_2^{\delta-1}}\, x_2 = w \quad\Longleftrightarrow\quad x_2(p,w) = \frac{w}{p_1^\delta + p_2^\delta}\, p_2^{\delta-1}.$$

Similarly, it can be derived that

$$x_1(p,w) = \frac{w}{p_1^\delta + p_2^\delta}\, p_1^{\delta-1}.$$